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Formation and Characterization of Poly(acrylic acid) on Silica Particles Irradiated by γ-Ray Radiation

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Abstract: Organic/inorganic hybrid gels were directly prepared by polymerization on the peroxide surface of silica particles where the particle surface was irradiated by a ^{60}Co γ-ray. These hydrogels have no residues of initiators or cross-linkers, so they can be used in biocompatible gel applications. Wide Raman spectroscopy was used to verify the interaction between the particles and poly(acrylic acid) (PAA). We observed that covalent bonds existing between the peroxide particles and acrylic acid, and the hydrogen bonds between the acrylic acids. For these studies, we prepared hydrogels by varying the particles' concentration and the size of the silica particles to classify the number of reaction sites, which are the dominant factor for the chemical reaction between the silica particles and PAA.

Keywords: hybrid gel, hydrogel, γ-ray, poly(acrylic acid) gel, organic/inorganic hybrid.

Introduction

Organic/inorganic hybrid gels have gained considerable attention in recent years.[1-6] They have many applications which require sensitive exterior conditions including various solvents, pH, and temperature.[7,8] Previously, we reported a new way of synthesizing hydrogels having silica particles and poly(acrylic acid) (PAA) with a well-defined network structure and high mechanical strength.[9] To synthesize hydrogels by free radical polymerization, γ-ray radiation was used to activate the surface of the silica particles, which allows for the polymer chemical reaction at relatively low temperatures. The proposed mechanism for the formation of peroxide and the initiation of polymerization, as well as for the formation of a hydrogel, is provided in Figure 1.

In the mechanism, the surface of the silica particles acts as an initiator as ^{60}Co γ-rays produce oxygen radicals on the surface.[9,10] The oxygen radicals on the silica particles start to propagate and are terminated by the monomer, in this case, acrylic acid. The physical properties of this system, including the tensile strength of the hybrid gel, have been reported, but the chemical structure between the particles and polymer has not been reported.

As their cross-linking sites are irregularly distributed and the polymer chains have different lengths, their influence on the physical and mechanical properties of the gel is statistically distributed. As such, the connectivity between the particles and polymer chains can be reasonably studied by spectroscopic techniques. Recently, wide Raman techniques were employed for the investigation of non-ergodic systems like gels. Spectroscopic analyses thereby provide insight into the macroscopic binding properties of the hydrogel between the particles and the polymer chains. In this paper, we specifically sought to verify the existence of covalent bonds between the silica particles and PAA and the intermolecular hydrogen bonds between the PAAs. We also sought to explain how the mechanical properties are affected by these chemical bonds.

Experimental

Reagents. Tetraethyl orthosilicate (TEOS), ethanol, ammonia solution (NH_4OH), acrylic acid (AA) are purchased from Aldrich and used without further purification. Milli-Q (Millipore, USA) system was used to prepare distilled and deionized water (DI-water, resistivity of >18.2 MΩ-cm).

Synthesis of Silica Particles: Silica particles were synthesized by the well-known Stöber method.[11] A solution of ethanol 150 mL, TESO 9 mL, and NH_4OH 3 mL in a 250 mL volumetric flask was stirred for 24 h for the preparation of 70 nm radius particles. Silica particles of 120 and 260 nm radius were also prepared by varying the ripening time. The solution was washed three times in ethanol and two times in DI-water.

Synthesis of Organic/Inorganic Hydrogels. We prepared a 2 wt% solution of silica particles in water and irradiated it with ^{60}Co γ-rays (dose rate: 10 kGy/h) at KAERI, Jeongeup, Korea, under ambient conditions for 2 h to create

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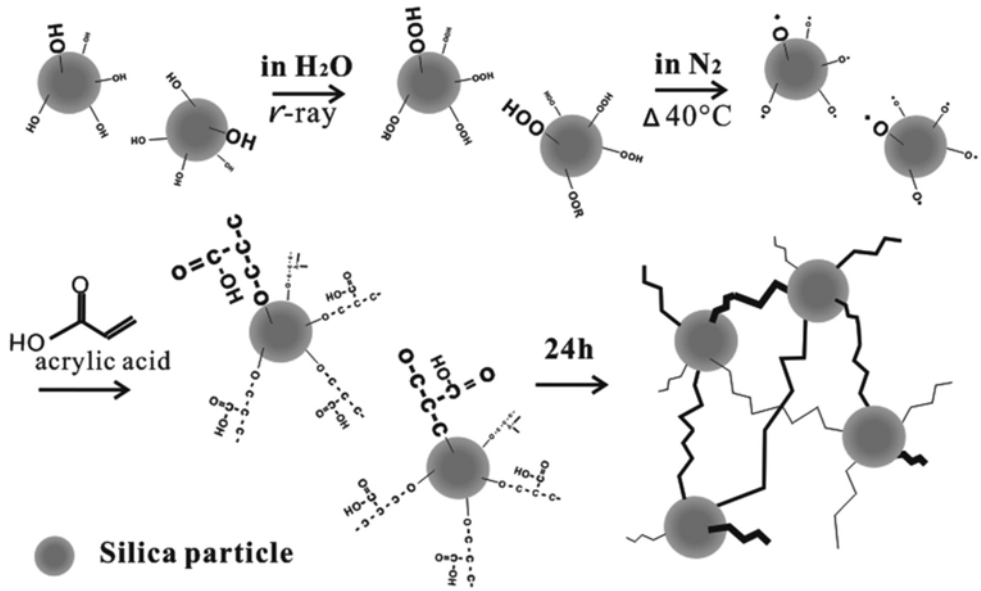

Figure 1. Mechanism for the formation of an inorganic/organic hydrogel.

peroxides on the surface of the particles. It is known that irradiation of oxygen in water results in the formation of peroxides on silica particles. With increasing temperature (40 °C), peroxides becomes the oxide radicals, which act as initiators and cross-linkers. A cylindrical glass ampoule was constructed for the synthesis of the hydrogel. The 2.5 mL of the peroxide silica particle solution (2 wt%) and 2 mL acrylic acid (monomer, AA) were mixed, and were allowed to react for 24 h. We used three cycles of the freeze-pump-thaw (freeze, evacuate volume *via* vacuum, thaw) method to remove air in the ampoule. Because the oxide radicals on the silica particle surface are very active and unstable, radicals easily disappear by other reactions. Finally, a well-defined network hydrogel was synthesized. To purify the synthesized hydrogel, we removed the non-reacted monomer and remaining water by placing in a freeze dryer for 48 h.

Instruments. The WAI scheme,[12] also called wide Raman spectroscopy, used in this study was developed by Kaiser Optical Systems (Ann Arbor, MI, USA). An excitation laser (785 nm, diode laser) was magnified to form a circular illumination area with a diameter of 6 mm (area 28.3 mm^2), so as to cover a large sample area for the non-ergodic system. Raman spectra were collected using an exposure time of 3 s over 5 scans. Particles and gels were observed by scanning electron microscopy (SEM) (JEOL, JSM-840A). Silica particles in DI-water was dropped at silicon wafer and dried at ambient condition. Then, we directly observed the particles by SEM at 15 kV. Silica particle embedded hydrogels were frozen in liquid nitrogen and then sliced quickly. A sliced hydrogel was attached on silicon wafer with silver paste, and then we observed the gel by SEM operated at 10 kV. Dynamic light scattering (DLS) to determine the particle size was performed using an ALV 5000/E/WIN multiple tau correlator that was applied by vertically polarized laser light (wavelength 632.8 nm) with a JDS-Uniphase He-Ne ion laser of λ_0=632.8 nm at room temperature.

Results and Discussion

Synthesis of Organic/Inorganic Hydrogel. The Stöber method can be used to easily control the size of silica particles. We made three silica particles which have radii of 70±10, 120±18, and 260±30 nm. The size of synthesized silica particles was confirmed by DLS, and the shape of the particle was determined by SEM. Figure 2 shows the SEM images of each particle.

After irradiation of γ-ray, the formation of peroxides on the particles was confirmed using iodometry,[13] which is a common method used to verify peroxide formation. Potassium iodide and isopropyl alcohol were added to the irradiated silica particle solution during heating and refluxing for 30 min. The solution gradually changed to a yellow color, indicating the formation of I_2, and thereby establishing the presence of peroxides. Changes in the color of solution upon addition of IPA and KI to the solution of Si particle are presented in the supporting information. Figure 3 shows the SEM images of hydrogels prepared with silica particles. Figure 3(a) is the surface image of the hydrogel, and Figure 3(b) is a cross-sectional image of the hydrogel after cutting the gel. The distance between two neighboring silica particles varied with the concentration of silica particle in the solution and the size of silica particle. The average distance between the particles was calculated from the concentration of silica particle and the size of the gel. In the case of the gel

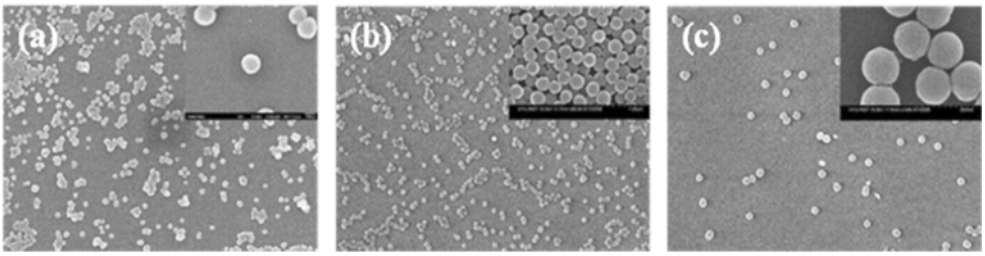

Figure 2. SEM images of the synthesized silica particles by Stöber method. (a) 70 nm radius of silica particle, (b) 120 nm radius of silica particle, and (c) 260 nm radius of silica particle.

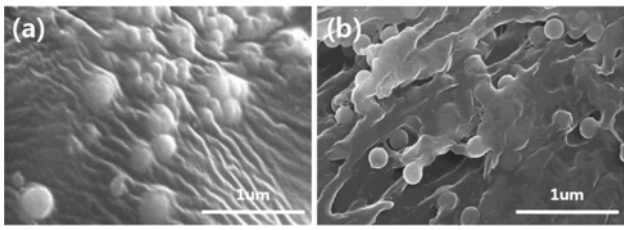

Figure 3. SEM images of synthesized hydrogel made from silica particles and poly(acrylic acid). (a) SEM image of the surface of the synthesized hydrogel, and (b) SEM image of the inside of the synthesized hydrogel.

which has 2 wt% 240 nm particles, the average distance between the particles was ~860 nm.

It is possible for two neighboring silica particles to be joined by a grafted chain, either by the mutual termination of growing chains that initiate at two silica particles, or by the termination of a grafted chain from one silica particle. In general, entanglement coupling is not likely to cause a significant enhancement in strength or toughness within a highly swollen gel because the chains are highly mobile and can slip past the entanglements. However, PAA chains are known to form inter- and intra-molecular hydrogen bonds that act as sticking points and enhance the dissipation involved in pulling out the entanglements.[14] This assumption has been proven by the fact that the synthesized hydrogel is easily swollen in a basic solution to suppress hydrogen bonding and cannot be dissolved in a low pH solution. (See the supplement information.) These gels are also highly elastic.[9]

Chemical and Physical Bonding on Organic/Inorganic Hydrogels. The polymerization of the hydrogel was confirmed by monitoring the conversion of the C=C bond of acrylic acid into a saturated C-C linkage by Raman spectroscopy. Silica particles could be joined by a grafted chain, and the entanglement of PAA chains connecting silica particles is likely to cause significant strength or toughness enhancement within a gel. Raman spectra of the hydrogel show the bonding structure of the silica particle itself and PAA, which has Si-O-C bonding. Figures 4-6 show the Raman spectra of the prepared hydrogels, which have different sizes of embedded silica particles and different solution concentrations. We try to maintain concentrations of the particles

Table I. Assignment of Raman Spectrum for the Silica Particle/PAA Hybrid Gel

Functional Group	Wavenumber (cm^{-1})
Si-O-C Stretching	840
C-COOH Stretching	855
O-H Out of Plane Bending	910
CH_2 Rocking	1010
C-CH_2 Stretching	1080
CH_2 Twisting	1290
C-O Stretching	1410
CH_2 Deformation	1460
C=C Stretching	1635
C=O Stretching	1720

and to see the size effect by adjust the baseline of the peaks. Peak assignments of the hydrogel are summarized in Table I.

The Si-O-C symmetric bond is observed at 840 cm^{-1}.[15] The other functional groups are in the range from 900 to 1800 cm^{-1}, and include O-H bending (910 cm^{-1}), CH_2 rocking (1010 cm^{-1}), C-CH_2 stretching (1080 cm^{-1}), CH_2 twisting (1290 cm^{-1}), and C-O stretching (1410 cm^{-1}), CH_2 bending (1460 cm^{-1}), C=O bond (1720 cm^{-1}).[16] Figure 4(a) presents the PAA itself and 3 different sizes of silica particles in the hydrogel. Figure 4(b) shows the comparison between the PAA polymer itself and the gel, which has 70 nm silica particles. Distinct differences are near 850 and 1600~1800 cm^{-1}, therefore details in these regions are provided in Figure 4(b) and (c).

The hydrogel has characteristic peak at ~850 cm^{-1}, which represents a combination band of Si-O-C and the PAA chain. Figure 5 shows the Raman spectra near 850 cm^{-1} for the gel which has the same size, 70 nm, but different particle concentrations. Figure 5(a) is 0.1 wt% Si particles and (b) is 2 wt% particles, which means (b) has more reaction sites than that of (a). Si-O-C contributes the peak at 840 cm^{-1} and PAA contributes at 855 cm^{-1}. Figure 5 also shows the fitting curves, which illustrates the two different contributions of the Raman peaks. For increasing reaction sites, we observed an increase in the shoulder peak at 840 cm^{-1}, which

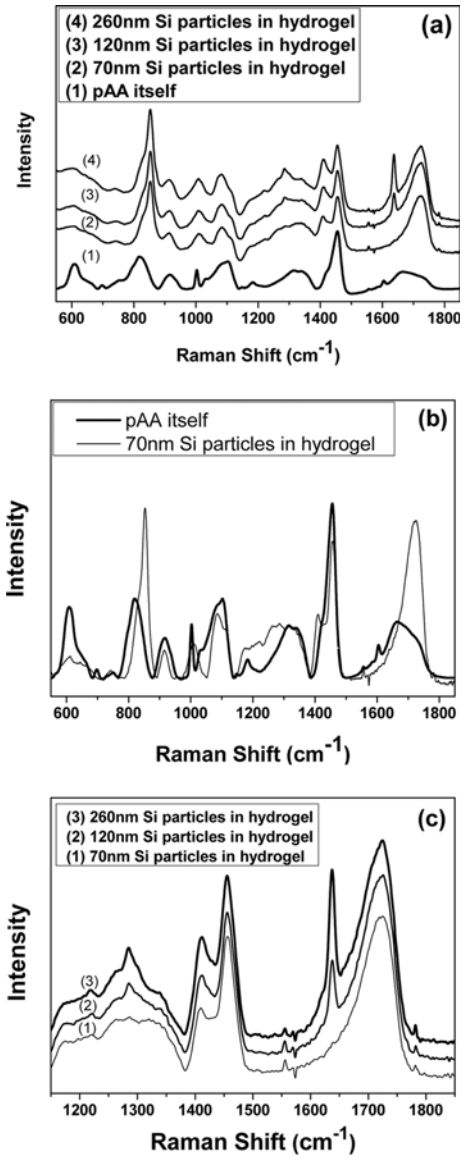

Figure 4. Raman spectra of (a) PAA itself, 70, 120, and 260 nm silica particles (2 wt%) in PAA gel, respectively. (b) Comparison between PAA itself and 70 nm silica particles in PAA hydrogel, and (c) characteristic peaks from 1200 to 1800 cm^{-1} for the 70, 120, and 260 nm silica particles in PAA gel.

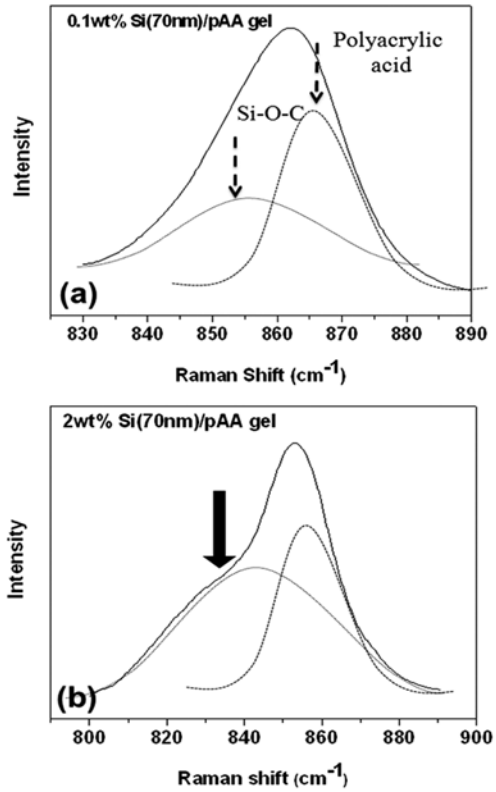

Figure 5. Raman spectra of the hydrogel made from acrylic acid with (a) 0.1 wt% silica particles 70 nm in size and (b) 2.0 wt% silica particles 70 nm in size.

indicates an increase in the Si-O-C contribution due to an increased number of sites. These observations demonstrate that covalent bonds increase between the Si particles and PAA by increasing the reaction sites.

The peak at 1635 cm^{-1} reflects the C=C stretching mode, while the one at 1290 cm^{-1} peak is the CH_2 twisting mode of acrylic acid. Hydrogels containing 260 nm Si particles have two sharp peaks at around 1635 and 1290 cm^{-1}.[17] We want to focus on the 1635 cm^{-1} peak which disappears with decreasing particle size in hydrogels. Smaller Si particles in the same concentration have more active surface area than those of larger silica particles. Thus, smaller SiO_2 particles have more peroxide groups than larger SiO_2 particles when irradiated by γ-ray radiation. So, the smaller particles embedded in the hydrogel result in more extensive connectivity. Finally, un-reacted acrylic acid monomer, which has sharp peak at 1635 cm^{-1}, remains inside the gel with larger particles. When the swelling and de-swelling process was repeated by changing the basic solution, a reduction in this 1635 cm^{-1} peak was observed.

The components of each carbonyl stretching in the Raman spectra were deconvoluted by a curve fitting program (Origin®), on the assumption that the peak shape of the components has a Gaussian distribution. The asymmetric band is due to C=O stretching mode centers near 1720 cm^{-1} and spans from 1600 to 1820 cm^{-1}, as shown in Figure 6 on an expanded scale. Figure 6 further shows the Raman spectra for samples having the same size of particles with different particle concentrations, (a) and (b), and different size of particles having the same concentration, (b) and (c). Samples with the same particle concentration and different particle sizes were also prepared to compare the bonding sites of the particles. For samples of 0.1 and 2 wt% of silica particles having the same particle size, the peak at ~1700 cm^{-1} increases for the high concentration sample. Therefore, the number of sites of C=O bonds in hydrogel containing 2 wt%

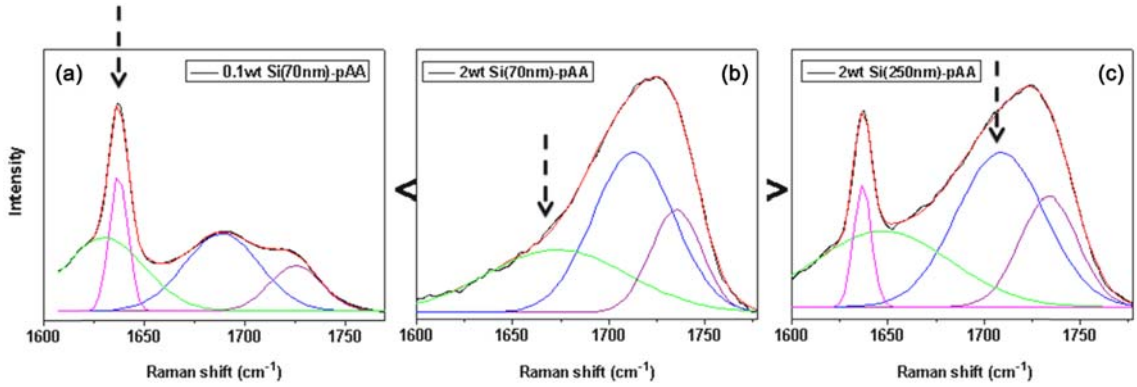

Figure 6. Raman spectra to show the silica particle size effect in hydrogels from acrylic acid, (a) hydrogel with 0.1 wt% of SiO_2 (r: 70 nm), (b) hydrogel with 2.0 wt% SiO_2 (r: 70 nm), and (c) hydrogel with 2.0 wt% SiO_2 (r: 250 nm). <, > indicate quantity of reaction sites.

Si particles is higher than in the hydrogel containing 0.1 wt% Si particles. There are at least three bands existing in the C=O stretching region, which are located at 1650, 1715, and 1740 cm^{-1},[18] respectively. Curve fitting, however, indicates four bands in the envelope, located at 1635, 1650, 1715, and 1740 cm^{-1}. The 1635 cm^{-1} band is assigned to a stretching vibration of the C=C bond of unreacted acrylic acid. The other bands at 1650, 1715, and 1740 cm^{-1} correspond to C=O stretching modes of free COOH, and terminal and inner COOH in the oligomeric forms, respectively.

C=C bonds disappear due to the increasing number of reaction sites from 0.1 to 2 wt% Si particles in the gel. Simultaneously, the contribution of oligomers C=O bonds is increased by increasing the reaction sites. At the same weight % for different particle sizes, the larger particles, which have fewer reaction sites, show large contributions of C=C bonds because fewer reactions go to completion. These results support that many double bonds are consumed on the surface of the particles.

Conclusions

We have successfully synthesized a smart hydrogel using various particle size and concentration with a unique microstructure. Different kinds of concentration and various sizes of silica particles were employed to investigate the bonding of hydrogel prepared by graft polymerization through γ-ray radiation. In this case, polymer chains are chemically grafted onto the surface of silica particles, which allows for control of the properties of each hydrogel by manipulation of the ratio of monomer to water and the size of the particles. In addition, it is possible to obtain new hydrogels with special swelling properties.

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Supporting Information: The iodine method to indicate the peroxide formation and the swelling behavior of hydrogel in a basic solution. These materials are available *via* the Internet at http://www/springer.com/13233.

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